JAMES MITCHELL

Research Engineer, Staff Level

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PROFESSIONAL SUMMARY

Accomplished Research Engineer with 14 years of experience designing and implementing scalable systems for data-intensive applications. Specialise in machine learning infrastructure, distributed computing, and system optimisation. Known for translating complex research concepts into production-ready solutions and mentoring junior engineers through technical challenges. Based in Oxford with a passion for solving architectural problems that bridge academic research and commercial deployment.

WORK EXPERIENCE

Staff Research Engineer

Temporal Systems Ltd (Previously known as ChromaFlow Analytics)

www.temporalsystems.co

Oxford, United Kingdom

January 2021 - Present

Architected and led development of a distributed time-series processing platform handling 2+ billion events daily across multiple data centres

Designed and implemented custom query optimisation engine that reduced average query latency by 38% for real-time analytics workloads

Established best practices for code review, testing, and deployment across engineering team of 8 researchers and engineers

Mentored 4 junior staff engineers on systems design principles, performance profiling, and debugging distributed systems issues

Senior Research Engineer

DataVault Technologies

www.datavault-tech.io

London, United Kingdom

March 2019 - December 2020

Led research initiatives exploring efficient data compression algorithms for genomic datasets, resulting in 2 peer-reviewed publications

Developed Python-based ETL pipeline processing 500GB+ of structured data daily from heterogeneous sources

Collaborated with external research institutions on grant-funded projects totalling GBP 1.2M in funding

Implemented automated monitoring and alerting system for 15 production microservices using open-source tools

Research Engineer

Oxford Computational Research Group

Department of Computer Science, University of Oxford

Oxford, United Kingdom

September 2010 - February 2019

Conducted research on machine learning model interpretability, publishing 6 peer-reviewed papers in top-tier conferences

Developed C++ simulation framework used by 12+ researchers across the department for comparative algorithm analysis

Built and maintained Linux cluster infrastructure supporting 200+ concurrent research jobs with resource scheduling system

Collaborated with industry partners including Microsoft Research and Google Brain on joint research initiatives

EDUCATION

Doctor of Philosophy (PhD) in Computer Science

University of Cambridge

Specialisation: Distributed Systems and Fault Tolerance

Graduated: 2010

Bachelor of Science (BSc) in Computer Science with First Class Honours

University of Oxford

Specialisation: Artificial Intelligence and Algorithms

Graduated: 2008

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, Go, Scala

Data Processing: Apache Spark, MapReduce, Kafka, Flink, Pandas, NumPy

Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost, statistical analysis

Distributed Systems: Kubernetes, Terraform, ArgoCD, Docker, service mesh architecture

Databases: PostgreSQL, MongoDB, Cassandra, Redis, DynamoDB

Cloud Platforms: AWS, Google Cloud Platform, Azure

Development Tools: Git, Jenkins, Grafana, ELK stack, JIRA

Research Methodologies: Experimental design, statistical testing, literature review, paper writing

ADDITIONAL INFORMATION

Publications: 8 peer-reviewed papers in venues including ICML, NeurIPS, and Systems conferences

Certifications: AWS Certified Solutions Architect (Professional), Google Cloud Professional Data Engineer

Languages: English (Native), French (Conversational)

Active contributor to open-source projects in data systems space